Digital Signal Encoder/Decoder: Specification Report
ITT 036 Programming Assignment 1 | Autumn 2026

1 Overview

A C++17 application for digital signal generation, line encoding, scrambling, and waveform visualization, with PCM encoding and decoding.

- **Digital mode:** Random and constrained-zero bitstreams; longest palindromic substring; NRZ-L, NRZ-I, Manchester, Differential Manchester, AMI, B8ZS, and HDB3.
- **Analog mode:** Waveform sampling, PCM encoding/decoding, and reconstructed signal visualization.

2 Language, Libraries and Tools

- **Language:** C++17
- **Graphics:** SFML
- **Build:** CMake
- **Version control:** Git/GitHub

3 Assumptions and Conventions

- **Levels:** Positive = +1, negative = −1, zero = 0.
- **NRZ-L:** 1 → +1; 0 → −1.
- **NRZ-I:** 1 inverts the level; 0 keeps it. Initial level: +1.
- **Manchester (IEEE 802.3):** 1 = (−1, +1); 0 = (+1, −1), with a mid-bit transition.
- **Differential Manchester:** Always transitions mid-bit; 0 also transitions at the bit start. Initial level: +1.
- **AMI:** 0 → 0; successive 1s alternate between +1 and −1. Previous polarity starts at −1.
- **B8ZS/HDB3:** Applied to AMI; replace eight/four consecutive zeros using the respective bipolar-violation rules.
- **PCM:** Quantization levels depend on bit depth; codewords and reconstruction follow the implementation.
- **Generator:** Supports random bits and runs of four/eight consecutive zeros.
- **Palindrome:** The longest contiguous palindromic substring is returned, not a subsequence.

4 Build, Run and Test

From the project root:

- **Configure:** `cmake -S . -B build`
- **Build:** `cmake -build build -j1`
- **Run:** `./build/signal-codec`
- **Test:** Run available tests for encoding, scrambling, PCM, and edge cases.

Requires a C++17 compiler, CMake, and SFML development libraries.

5 References

- ITT 036 Programming Assignment 1 specification and course notes.
- Manacher's algorithm: `https://cp-algorithms.com/string/manacher.html`
- SFML: `https://www.sfml-dev.org/`
- CMake: `https://cmake.org/documentation/`

6 Acknowledgment

ChatGPT assisted with concept understanding, implementation planning, debugging, and documentation.